of the enzyme so that the affinity of the substrate for the active site is reduced.

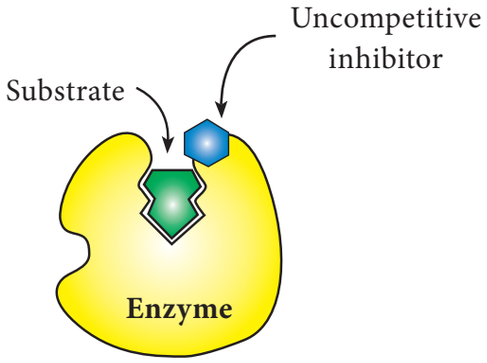

Figure 4.10 Uncompetitive inhibition

Table 4.2 Compersion between competitive and Noncompetitive inhibition

Sl. No.	Competitive inhibition	Noncompetitive inhibition
1.	Inhibitor resembles the substrate	Inhibitor has no structural resemblance with the substrate.
2.	Inhibitor binds at the active site.	Inhibitor binds at a site other than active site
3.	Enzyme binds either with substrate or inhibitor	Enzyme binds with both substrate and inhibitor
4.	Reversible	Reversible
5.	Can be overcome by increasing substrate concentration.	Cannot be overcome by increasing substrate concentration

4.6 Industrial applications of enzymes

Enzymes are widely used in food, pharmaceutical and chemical industries. Bacterial enzymes are used for the fermentation of food items. For example,

i. Making of curd from milk by *lactobacillus acidophilus*.

ii. Producing yoghurt and cheese from milk by *streptococcus thermophilus*.

iii. Fermenting rice and blackgram by *leucanostoc mesenteroides* for preparing delicious idlies.

iv. In washing powders enzymes are incorporated to remove stains from clothes.

v. Fructose syrup is produced by the isomerization of glucose by glucose isomerase.

vi. Penicillin acylase is used to convert penicillin to semisynthetic penicillins.

vii. Glucose and galactose are produced from the discarded portion during cheese making by lactase enzyme

viii. Enzymes are used for desizing fabrics. This kind of enzymatic desizing does not weaken the fabrics.

ix. In the manufacture of leather, the hide is made free from hair. This is done by employing pancreatic enzymes.

x. Pepsin is used to digest gelatin in the process of recovering silver from photographic films.

4.7 Medical applications of enzymes

i. Streptokinase or irokinase is sometimes used to lyse intravascular blood clots.

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